



OPEN Network-based intrusion detection using deep learning technique

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A high growth rate in network traffic and the complexity of cyber threats have made it necessary to create more effective and flexible intrusion detection systems. Most traditional Network-based Intrusion Detection Systems (NIDS) can become weak at detecting new patterns of attacks due to the use of obsolete data or traditional machine learning models. To overcome the mentioned constraints, the current research presents a new deep learning solution that combines Sequential Deep Neural Networks (DNN) and Rectified Linear Unit (ReLU) activation unit with an Extra Tree Classifier feature selection procedure. The proposed model is trained and tested on the new rich and up-to-date UNSW-NB15 set, which provides a realistic reflection of the real-life network traffic and attack vectors. The interesting novelty of this study is the tactical use of ReLU-based DNN combined with feature optimization through the Extra Tree Classifier, which not only overcomes general problems like vanishing gradients and overfitting but also greatly increases the interpretability of the model and the efficiency of its computation. This dimensional reduction of the feature space (43 to only 8 highly relevant features) retains the high accuracy of the model but with better inference speed, which is a crucial aspect of the real-time deployment of NIDS. The results show that with the Sequential DNN approach, the binary class (0 for normal and 1 for attack records) achieved 97.93% accuracy, 97% Precision, 97% Recall and 97% F1-score. Furthermore, the detailed experimental testing, such as ROC curves and Confusion Matrices, confirmed that the Sequential DNN performed well in comparison to other Existing Studies. These findings underscore the effectiveness of deep learning architectures enhanced with optimized feature selection in detecting network intrusions, making the proposed system a promising solution for securing critical infrastructure in sectors such as finance, healthcare, and government networks.

Keywords Network attacks, Network-based intrusion detection system (NIDS), Deep neural networks, Extra tree classifier, Activation function

Due to cyber-attacks that are becoming more common and advanced, IDS plays a vital role in contemporary cyber security systems because organizations require effective ways to monitor their networks and protect themselves against all kinds of cyber-attacks^{1,2}. Conventional security solutions like firewalls are ineffective at identifying and stopping sophisticated threats, particularly zero-day attacks and abnormal activities. This brings an imperative necessity in the development of smart solutions such as IDS that will examine network traffic patterns and detect malicious behaviors on a real-time basis. This study aims to make use of Deep Learning (DL) techniques because of their demonstrated effectiveness at analyzing large amounts of data via high-dimensional, high-complexity data such as network traffic logs. Compared with classical machine learning-based models that require extensive exploitation of experts to engineer features, deep learning models can learn a hierarchical feature representation of raw data, which permits more precise and semi-dynamic intrusion identifications.

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Additionally, DL models, particularly Deep Neural Networks (DNNs), present higher performances in detecting attacks with low and previously unknown patterns because of their non-linearity learning abilities and resilience. To support these motives, we provide a new framework, which is a combination of a Sequential Deep Neural Network and ReLU activation functions and Extra Tree-based feature selection with the modern UNSW-NB15 record. This method is a response to computation efficiency and detection performance, increasing performance metrics compared to the benchmark models. There are two types of Intrusion Detection Systems: Host-based and Network-based. A Host-based IDS (HIDS) is a security tool often deployed throughout a network and installed on individual hosts or endpoints. It monitors hosts' activities and identifies potentially harmful or suspicious behavior. The operation of HIDS software involves tracking the action on the host device itself to search for indications of unauthorized access, malicious code, or other security concerns. For instance, if a user attempts to access a file for which they do not have permission, the HIDS software can alert security staff to investigate and respond accordingly. A HIDS is a powerful tool for detecting and responding to security risks on individual hosts or endpoints, and it can help companies enhance their overall security posture. A Network-based IDS (NIDS) that monitors and evaluates network traffic is a security tool that assists in identifying possible security issues and taking the necessary steps to fix them. The operation of NIDS software involves monitoring network data and then producing warnings or taking measures in response to any possible vulnerabilities. A wide range of security risks, including malware, DoS attacks, and other types of malicious activity, can be detected by NIDS systems. Figure 1 represents a network security system presenting both a host-based Intrusion Detection System (HIDS) and a network-based IDS. Presenting the setup of firewalls, a router, and a WWW server for managing, monitoring and controlling network traffic for security reasons.

Likewise, in recent times, research related to deep learning-based Intrusion Detection has been carried out using different datasets such as KDDCUP99, KDD98, and NSL-KDD⁴. Nonetheless, these datasets influence the results of intrusion systems due to the absence of modern-day traffic data. Henceforth, enhancing the functionality of the current NIDS feels the necessity of adapting modern and state-of-the-art datasets. In this research work, a UNSW-NB15⁵ dataset-based framework has been developed for the detection of attacks in the network by using a deep learning technique^{6,7}. Here are the key novelties of research:

- Propose a new combination of a Sequential Deep Neural Network (DNN) with Rectified Linear Unit (ReLU) activation and an Extra Tree Classifier feature selection to the UNSW-NB15 dataset. Previous works have considered DNNs but with conventional activation functions, such as sigmoid or tanh, which have the drawback of the vanishing gradients problem and poor behavior with deeper networks.
- Apply the Relu Activation function to mitigate vanishing gradients problems.
- Apply Extra Tree Classifier as a Feature Selection Technique: Dimensionality reduced to 8 important features out of 43 features, enhancing the computation capability without compromising accuracy.
- Achieved high-performance metrics in binary classification, including an accuracy of 97.93%, along with 97% precision, recall, and F1-score, surpassing previous benchmark studies.

The structure of the paper is as follows: The related works that are similar to the current study are presented in Section II. Section III discusses the methodology for the development of the framework. The results and discussion are presented in Section IV. While in the end, the conclusions are presented in Section V.

Related work

Deep learning-based intrusion detection systems have been the subject of extensive analytics research over the last few years. Due to its capacity for growth and learning, this field of study is gaining popularity. Because of this, it is very successful and highly effective in addressing an alarming rise in unknown attacks. The use of a deep learning approach on the KDD-99 dataset to create a deep learning method for finding anomalies has been analyzed by^{8–11}. Since the UNSW-NB15 data set is complicated¹², so according to the experimental findings, UNSW-NB15 was more complicated than the dataset of KDD99 and is now regarded as a new benchmark data set to evaluate NIDS. In¹³, an improved way of using vision-based deep learning models to process network

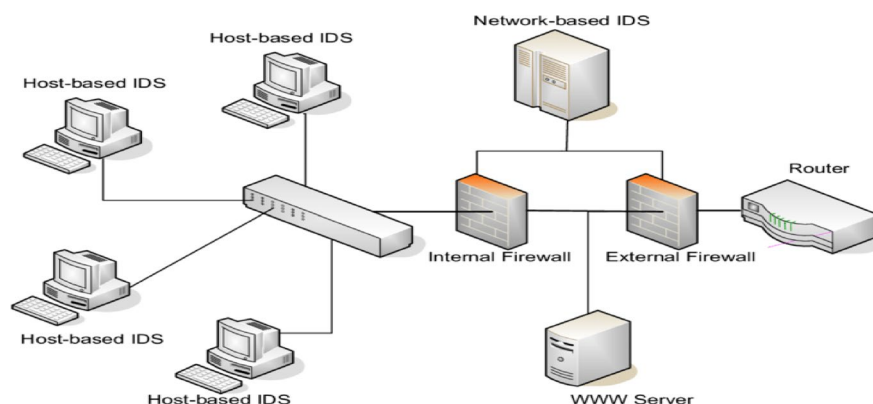


Fig. 1. Host-based IDS and network-based IDS³.

intrusion detection systems is presented. While a multiclass Artificial Neural Network (ANN) for intrusion detection in computer networks has been introduced¹⁴. Whereas, a deep neural network approach was evaluated on the datasets of KDDCUP 1999 and UNSW-NB15. The results of tests show that the model may be able to find cyberattacks in computer networks. Similarly, the application and analysis of DNN as a deep learning model to identify and categorize an adaptive and efficient intrusion detection system has been carried out¹⁵. While, a two-stage classifier based on a subset of protocols and the RepTree algorithm for network IDS has been presented in¹⁶. The datasets of UNSW-NB15 and the NSL-KDD were used in the study to test the effectiveness of the classifier. The results of the experiments showed that the detection accuracy on the complete datasets of UNSW-NB15 and NSL-KDD was 89.85% and 88.95%, respectively. In¹⁷, a two-stage strategy for network intrusion recognition was proposed. The approach was founded on the GoogleNet Inception model and the deep convolution organization (CNN) model.

According to the results, the strategy that was suggested for enhancing the consistent quality of NIDS was effectively presented. Using the dataset from NSL-KDD¹⁸, analyzed three different algorithms of machine learning. In the study, the NSL-KDD dataset was pre-processed, the selection of ML classifiers (SVM, RF, and ELM) was carried out, and accuracy, precision, and recall were measured. In¹⁹, a dataset from KDDCUP99 was utilized. The study proposed a decision tree-based IDS framework in a big data over fog environment. The findings demonstrated that the suggested approach was both reliable and efficient. While²⁰ examined the benefits and drawbacks of contemporary machine and deep learning intrusion detection algorithms. The study²¹ demonstrates an efficient incentive mechanism in Vehicular Networks that is simple but helps to select the agents having high-quality resources and to encourage them to act in the learning process so that the resulting aggregated model would be accurate, and the network overhead would be smaller. The paper²² introduces a recently identified front-end Web attack that leverages the existing object storage services vulnerability of cloud providers to evade content security policy. The proposal of the study²³ is to introduce an unpacking system to the smart blockchain applications: AutoD, which is founded on the deception call on the JNI layer of Android ART. The study²⁴ propose a lightweight and simple to implement and augment unpacking framework named DeepAutoD (hereinafter referred to as d-ad) on the safety of distributed computers. The authors of the study²⁵ suggests an automated and consistent methodology of detecting BGP community attributes in AS-level. The propose study²⁶ an interactive gradient shielding (IGS) approach that first chooses sensitive areas and then uses gradient-based attack.

The article²⁷ suggests a new Network-based Intrusion Detection Machine Learning (NIDML) model, which is a learning model consisting of an ensemble of Decision Trees, Random Forests, K-nearest neighbors, Neural Networks, and Ensemble methods. The LUFLOW dataset on which each of the algorithms was trained and tested ranges between 93.03% and 99.96% accuracy of the resulting models. In the paper²⁸, modern technical methods of improving the quality of work of intrusion detection systems are studied due to the critical importance of ensuring the security of computer networks against malicious activity. The model trains and validates the DNN-based IDS on the NSL-KDD dataset, which is a common benchmark in IDS research. The model trained and validates well with an accuracy of 91.30% and 94.38%, respectively. In this research²⁹, a Hybrid Deep-Learning-Based Network Intrusion Detection System (HDLNIDS) is used to build a hybrid intrusion detection system based on deep-learning to identify attacks across a network. To identify the effectiveness of the proposed system, the experiments are performed on publicly available benchmark CICIDS-2018 data. Simulation results of the research prove that the suggested HDLNIDS is more effective than the existing intrusion detection methods, having an average accuracy of 98.90% in identifying malicious attacks. The study³⁰ presents a deep learning-based detection framework to enhance the accuracy of threat detection by automatically extracting features and modeling nonlinearity. A detection model based on KDD Cup 1999 and UNSW-NB15 data sets supporting convolutional neural network (CNN), long short-term memory network (LSTM) and hybrid architecture designed to efficiently optimize preprocessing of data and design lightweight models in a systematic manner. It has been observed through experiments that the overall performance of the proposed CNN- LSTM integration model on the two types of datasets is much higher than the traditional machine learning and support vector machine models. F1 value is 0.9 (90%) of detecting KDD Cup 1999 data set and 0.86 (87%) of detecting UNSW-NB15 data set. The study³¹ suggests a combined strategy employing the convolution neural network (CNN) and the long short-term memory (LSTM) with the hyper parameters optimization based on Bayesian optimization method in attacks detection in IoT. A novel stacked auto-encoder and soft-max classifier-based two-stage deep learning model for efficient network intrusion detection was proposed by³². Whereas the author³³ used the dataset of UNSW-NB15 as part of a two-step process for finding network intrusions based on anomalies. The researchers selected the dataset attributes that were most useful for ML using a variety of techniques, some of which include Recursive Feature Elimination and Random Forests. The implementation of a method for deep learning to automatically detect anomalies was studied in³⁴. The experiment results demonstrated that the newly proposed deep learning model was better than the previous model in terms of performance.

A comparison of the performance of classical machine learning techniques was carried out in³⁵, which required significant feature engineering on the dataset of NSL-KDD. The research conducted using deep and machine learning-based network intrusion systems is still not sufficient. However, some of the related research work is summarized in Table 1.

Reference	Dataset (Complete/Partial)	Model	Accuracy	Research Gap
12	UNSW-NB15 (Partial)	ANN	Accuracy between 78.47% and 85.56%	Techniques for data Pre-processing have not been used in this research paper to improve accuracy further.
36	UNSW-NB15	ANN	Accuracy between 63.9% and 99.32%	The model is implemented on Binary Classification. The 80,000 data is selected for research.
11	UNSW-NB15	ANN	Accuracy: 92%	A good Binary classification accuracy has not been achieved due to class imbalance.
32	UNSW-NB15	Deep-stacked auto-encoder (DSAE) neural network	Accuracy: 89.13%	The paper does not demonstrate how well the model recognizes minority attack classes.
37	UNSW-NB15	ANN	Accuracy between 53.43% and 82.66%	Due to a problem with class imbalance, the proposed model has not been able to attain good accuracy.
38	UNSW-NB15	Random Forest	Accuracy: 87.09%	The paper applies the Random Forest model to binary classes. The experiments concluded that a Random Forest cannot achieve good accuracy.
27	LUFlow dataset	Decision Trees, Random Forests, K-nearest neighbors, Neural Networks, and Ensemble methods	Accuracy: 99.96%	LUFlow dependency; no cross-dataset validation; unproven class balancing
29	CICIDS-2018	HDLNIDS	Accuracy: 98.90%	No reporting of inference speed (critical for real-time NIDS).
28	NSL-KDD	Deep Neural Network	Accuracy: 94.38%	Old Dataset
6	UNSW-NB15	Random Forest	Accuracy: 95.30%	Minimal Discussion on Class Imbalance
7	UNSW-NB15	LSTM	Accuracy: 92.2%	Reliance on Specific Dataset Features

Table 1. Summary of existing research Work.

The brief assessment of related work makes it clear that more research is needed to pinpoint the characteristics of network attacks. Henceforth, finding a general model with greater accuracy for the attacks in the dataset is necessary. In this work, the use of a sequential deep learning model with an extra tree classifier has been carried out to detect intrusion. Moreover, a dataset called UNSW-NB15 has been used. Furthermore, there has been limited research that has used DNN to detect intrusion using the under-consideration dataset.

Methodology

The development of an efficient deep-learning model for the identification of attacks within a dataset is of high significance. Therefore, this study is focused on utilizing DNN, a deep learning algorithm, to detect attacks considering the UNSW-NB15 dataset. The overall framework representing the proposed methodology is depicted in Fig. 2. The steps involved in the implementation of the DNN model in the NIDS are discussed below.

The following flowchart 2 shows the procedure involved in predicting the intruders based on the UNSW-NB15 dataset. Pre-processing, such as standardization, feature selection, and splitting data into training and testing sets, was then followed by using an Extra Tree Classifier and a sequential deep neural network model for prediction.

Unsw-Nb15 dataset

UNSW-NB15 is a network intrusion dataset that has 44 distinct features. In this dataset, there are 43 input variables and one output variable. The last column of the dataset describes the binary-class feature. These variables were used in the training process of the DNN model.

Data standardization

Data standardization sometimes referred to as data scaling, is an important data pre-processing method used in a variety of data analysis and deep learning activities. In the proposed work, data standardization aims to convert a dataset's features into a common scale (between 0 and 1) to improve the accuracy of the model. The relationship used to achieve standardization is given as:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

Feature selection techniques

Feature selection is a method for choosing features that primarily correlate with and contribute to the dataset's targeted variable. There are different techniques for feature selection, such as information gain attributes, correlation attributes, and principle component analysis³⁹. However, in this framework, an extra tree classifier has been used. An extra tree classifier is a feature selection technique in which highly relevant features can be selected. After applying the feature selection technique, the data is ready for training the DNN model.

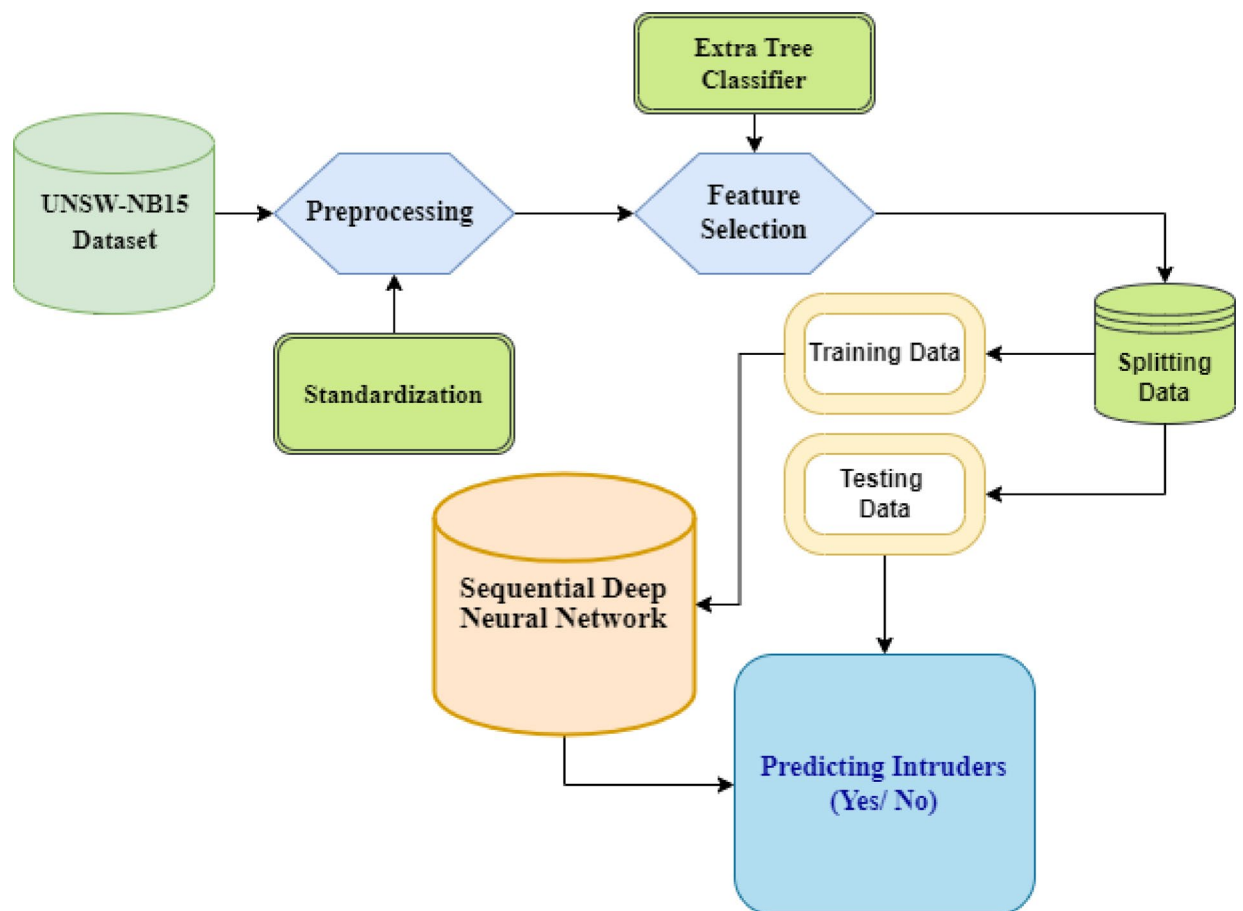


Fig. 2. Proposed model for network intrusion detection.

Deep neural network

An artificial neural network (ANN) that has several hidden layers in between the input and output layers is known as a deep neural network (DNN). Multiple neurons are present in each layer of a DNN, and these neurons are structured by one of three primary categories:

The input layer, which takes raw data or features from the dataset, is called the input layer. The hidden layer, with the presence of many hidden layers in a DNN, helps in evaluating the data input and learning to

Feature number	Feature/format	Feature number	Feature/format	Feature number	Feature/format
1.	Id(Integer)	16.	dloss(Integer)	31.	response_body_len(Integer)
2.	dur(Float)	17.	sinpkt(Float)	32.	ct_srv_src(Integer)
3.	proto(Categorical)	18.	dinpkt (Float)	33.	ct_state_ttl(Integer)
4.	service(Categorical)	19.	sjit (Float)	34.	ct_dst_ltm(Integer)
5.	state(Categorical)	20.	djit (Float)	35.	ct_src_dport_ltm(Integer)
6.	spkts(Integer)	21.	swin(Integer)	36.	ct_dst_sport_ltm(Integer)
7.	dpkts(Integer)	22.	stcpb(Integer)	37.	ct_dst_src_ltm(Integer)
8.	sbytes(Integer)	23.	dtcpb(Integer)	38.	is_ftp_login(Binary)
9.	dbytes(Integer)	24.	dwin(Integer)	39.	ct_ftp_cmd(Integer)
10.	rate(Float)	25.	tcprtt (Float)	40.	ct_flw_http_mthd(Integer)
11.	sttl(Integer)	26.	synack (Float)	41.	ct_src_ltm(Integer)
12.	dttl(Integer)	27.	ackdat (Float)	42.	ct_srv_dst(Integer)
13.	sload(Float)	28.	smean(Integer)	43.	is_sm_ips_ports(Binary)
14.	dload(Float)	29.	dmean(Integer)	44.	label(Binary)
15.	sloss(Integer)	30.	trans_depth(Integer)		

Table 2. Features of dataset.

correctly represent valuable patterns based on the information they are given. The mathematically hidden layer is described as:

$$z_j = \sigma \left(\sum_{i=1}^n w_{ji}x_i + b_j \right) \tag{2}$$

In the hidden layer, there are different types of activation functions, such as ReLU, Sigmoid, and Tanh functions. The ReLU activation function has been used in this framework to introduce non-linearity in the DNN model. The mathematical equation for ReLU is:

$$f(x) = \max(0, x) \tag{3}$$

- 1) Output Layer: following the processing of the information contained in the hidden layers, the output layer is responsible for producing the final predictions or outputs. The network's task determines the count of the output layer's neurons. In the case of a problem involving binary classification, for instance, there will be two output neurons reflecting the probabilities of belonging to each class, and in the case of a problem involving many classes, there will be several output neurons, one for each class. The mathematical equation for the output layer (binary classification) is.

$$\hat{y} = \sigma(Wx + b) \tag{4}$$

where:

- $\hat{y}$: is the predicted output (a probability between 0 and 1).
- σ : represents the sigmoid function, which is defined as: $\sigma(z) = 1 / (1 + \exp(-z))$.
- W : is the weight matrix connecting the last hidden layer to the output layer.
- x : is the output of the last hidden layer.
- b : is the bias term for the output layer.

The architectural diagram of Deep Neural Network is shown in Fig. 3.
The proposed work uses DNN models for binary class. In the binary class, there will be two output neurons. The following hyperparameters were used in the proposed model by using grid search strategy.

Hyperparameter	Value
Input Layer	8 neurons (based on 8 features selected by Extra Trees).
Hidden Layer	Three dense layers with 800, 800, and 400 units respectively.
Output Layer	Two neurons for binary classification
Optimizer	Adam
Learning Rate	0.001 (Default value for Adam)

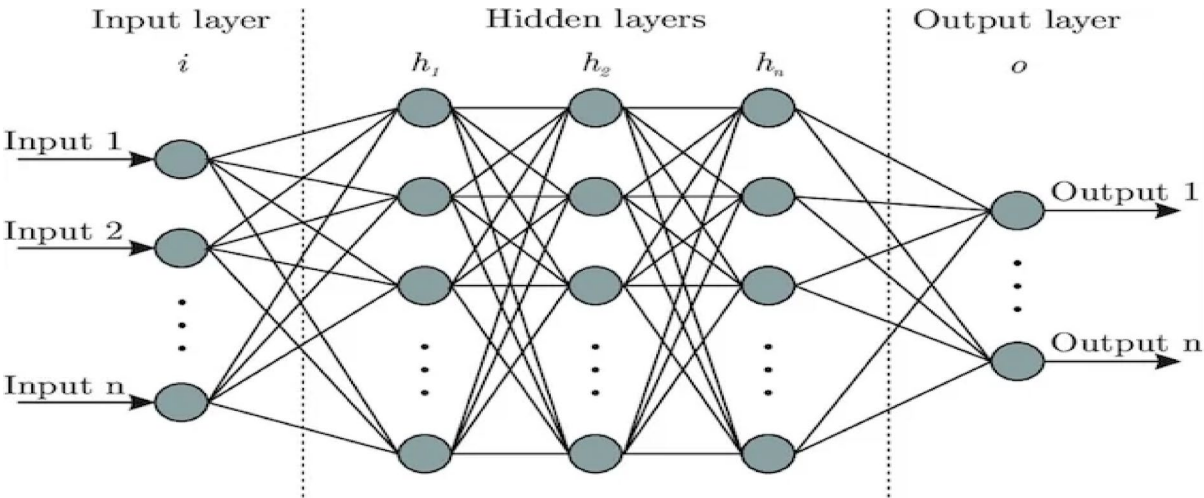


Fig. 3. Architectural diagram of deep neural network⁴⁰.

Hyperparameter	Value
Epochs	100
Batch Size	50
Loss Function	Categorical Crossentropy
Activation Function	Relu (use in Hidden Layer) Softmax (in the Output layer)

After performing the DNN model on the training dataset, the model will be predicted on the test dataset to evaluate the accuracy of the model. During the training of the model, certain issues and threats to the validity of the proposed work were observed, which are discussed in this section. A major issue during the training of the model was overfitting, which occurs when the model becomes highly specialized in learning from the training data and fails to generalize to new data. To resolve the problem of overfitting, regularization strategies such as L2 regularization, in conjunction with dropout, were utilized. Moreover, another major issue was data quality. To improve the data quality, careful pre-processing of the data, handling missing values, and removing duplicate records were carried out. During the training process of the proposed model, vanishing and exploding gradient issues were also faced, which were addressed by using an appropriate activation function, such as ReLU. Furthermore, hardware and computational limitations were also a concern, which degraded the computation of the proposed model. To solve this issue, hardware accelerators such as TPUs were used, whereas optimization of the code helped increase the speed of the training process.

Evaluation metrics

Some of the performance measures used to evaluate the models through the utilization of the confusion matrix are discussed below.

Accuracy shows the ratio of the number of data points that were correctly classified over the total number of data points⁴¹. Mathematically, accuracy is defined as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

Precision shows the ratio of true positive predictions as a percentage of all positive predictions. The following relationship is used for precision:

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

Recall estimates the number of predictions that came true out of all the positive instances⁴², where:

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

Finally, the F1-Score shows the harmonic mean of recall and precision⁴³. Mathematically,

$$F1 - score = \frac{2 * (Precision * Recall)}{(Precision + Recall)} \quad (8)$$

Results

The dataset considered for the current research is UNSW-NB15⁵, which has 175,341 records and is also imbalanced with a higher proportion of attack samples than normal traffic. To deal with the problem of class imbalance in the UNSW-NB15 dataset, the Synthetic Minority Oversampling Technique (SMOTE) is used in the pre-processing step. This method artificially increased the minority class by creating new realistic examples of the points that already existed so that the class distribution was balanced before the training of the model. Moreover, binary classification models are used on this dataset to evaluate the different metrics of the model. The binary classification model has two classes: Class 0 for no attack and Class 1 for attack. The dataset selected for training was 70%, and for testing was 30%.

The Pseudo-algorithmic of the proposed model is given below.

Input:

- X_train (feature matrix of training data)
- y_train (one-hot encoded labels)
- input_dim = 8 (number of input features)

Output:

- Compiled and trained classification model (model)

BEGIN1. **Initialize the model:**

model ← Sequential()

2. **Add input and hidden layers:**

model.add(Dense(units = 400, input_dim = 8, activation = 'relu'), Regularizer=L2)

model.add(Dense(units = 800, activation = 'relu'))

Dropout(0.2)

model.add(Dense(units = 800, activation = 'relu'))

Dropout(0.2)

model.add(Dense(units = 400, activation = 'relu'))

3. **Add output layer:**

model.add(Dense(units = 2, activation = 'softmax'))

4. **Compile the model:**

optimizer ← Adam()

loss ← categorical_crossentropy

metrics ← ['accuracy']

model.compile(optimizer, loss, metrics)

5. **Train the model:**

- Set training configuration:

batch_size ← 50

epochs ← 100

validation_split ← 0.33

- Train the model and store training history:

modelhistory ← model.fit(X_train, y_train, validation_split, batch_size, epochs)

END

Pseudo-algorithmic.

After taking the samples, an Extra Tree Classifier, which is a feature selection technique, is applied to different attributes of the dataset. The Extra Tree Classifier was not described in this research as an ordinary classifier but as a strong feature selection method that helped to select the most important features of the UNSW-NB15 data set. Since the dataset provided was of high dimension (43 features), feature reduction was extremely important to make the model not only computation-friendly but also accurate. As the Extra Tree Classifier is an ensemble-based algorithm, it measures the importance of the features against the average decrease in impurity acting upon the selected randomized decision trees. This is in order to weed out the characteristics that have little contribution to the ability to predict the model. The extra tree classifier selected only eight different attributes with the most highly relevant scores where the threshold value 0.021 was selected. All these attributes play a significant role in intrusion detection. As an example, sttl (source to destination time to live) and dttl (destination to source TTL) may indicate an anomaly in packet flow commonly related to spoofing or TTL expiration attacks. Connection-level statistics, such as the features ct_state_ttl and ct_srv_dst, are important to detect advanced patterns of misuse. The selection makes sure that only the most discriminative features are kept, which enhances the computation efficiency without substantial decreases in the accuracy of the model. The scores for this relevancy are summarized in Table 3.

S.No.	Selected attributes	High relevance values selected for attribute
1	Id	(0.1784)
2	State	(0.0372)
3	Sttl	(0.1897)
4	dttl	(0.1018)
5	dload	(0.0374)
6	swin	(0.0317)
7	ct_state_ttl	(0.0944)
8	ct_srv_dst	(0.0229)

Table 3. Selected attributes taken from the extra tree Classifier.

Epochs	Time	Loss	Accuracy	Val_accuracy
1	30s 18ms	0.1107	0.9590	0.9633
2	29s 18ms	0.0861	0.9682	0.9653
3	29s 18ms	0.0800	0.9700	0.9714
4	30s 18ms	0.0753	0.9713	0.9716
5	31s 19ms	0.0738	0.9715	0.9728

Table 4. First five selected rows of binary classification model.

Epochs	Time	Loss	Accuracy	Val_accuracy
96	28s 17ms	0.0515	0.9785	0.9783
97	28s 17ms	0.0514	0.9785	0.9800
98	29s 18ms	0.0517	0.9785	0.9813
99	32s 19ms	0.0508	0.9786	0.9804
100	31s 19ms	0.0516	0.9785	0.9818

Table 5. Last five selected rows of the binary classification model.

Class	Precision	Recall	F1-score	Support
0	0.99	0.91	0.95	16,721
1	0.96	0.99	0.98	35,882
Accuracy			97.93%	52,603
Macro Avg.	0.97	0.95	0.96	52,603
Weighted Avg	0.97	0.97	0.97	52,603

Table 6. Classification report of binary classification model.

After selecting highly relevant attributes, a standard scalar process is used which scales the selected attributes from the values 0 to 1. After applying the standardization process, the binary classification model is trained with 70% of the training data and 30% of the testing data. The model is trained with 100 epochs and a batch size of 50. As there are 100 epochs to fit the model and a description of all epochs cannot be shown in this limited space, the first five, Table 4, and the last five, Table 5, epochs are shown here.

As depicted in the last two columns of Tables 4 and 5, the trained accuracy (Accuracy) and test accuracy (Val_accuracy) are gradually increasing. While the loss values are decreasing for 0 to 99 epochs, as observed in Tables 4 and 5. The results show the true validation of the model fitting. To further confirm the strength and generalization ability of the proposed Sequential Deep Neural Network (DNN) model, the UNSW-NB15 data set was further subjected to 5-fold cross-validation. The findings revealed a standard deviation of (0.5 +/-) of the mean classification accuracy of 97.2%, showing that the model performed similarly on different data splits.

The results of binary classification in terms of accuracy, precision, recall, and f1-score are summarized in Table 6.

The table displays the findings taken from various evaluation matrices, which are required to verify the model's performance. The model performs well in terms of 97.93% accuracy.

The graph of the classification report is presented in Fig. 4.

The classification report metrics of a binary classification problem with precision, recall and F1 score of Class 0 and Class 1. Values are shown for each metric in both classes, and high performance in all metrics is shown for both classes.

Figures 5 and 6 present the learning curve accuracy and their loss drawn from Tables 4 and 5 for the binary classification model. The figure shows that as the no of epoch's increases, training and testing accuracy increases, and their loss is decreased.

To further validate the results, confusion matrices were used to classify attacks according to binary class, which is shown in Fig. 7.

Plot 4 shows the accuracy of a model during training and testing over 100 epochs. The blue line represents the training accuracy, and the orange line represents the testing accuracy. Both accuracies increase steadily, with the training accuracy reaching a higher value compared to the testing accuracy, which fluctuates somewhat during the training process.

Plot 5 shows the loss of a binary classification model for Network Intrusion Detection Systems (NIDS) over 100 epochs. The blue line represents the training loss, which decreases steadily, while the orange line represents the testing loss, which fluctuates but also tends to decrease. Both losses stabilize towards the end of the training process.

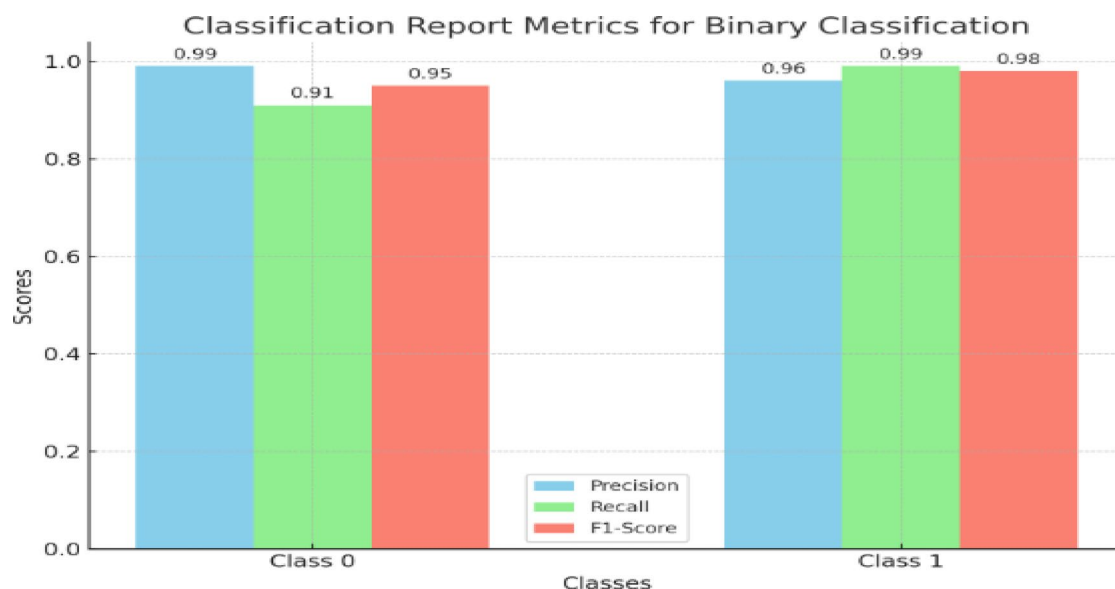


Fig. 4. Graph of classification report.

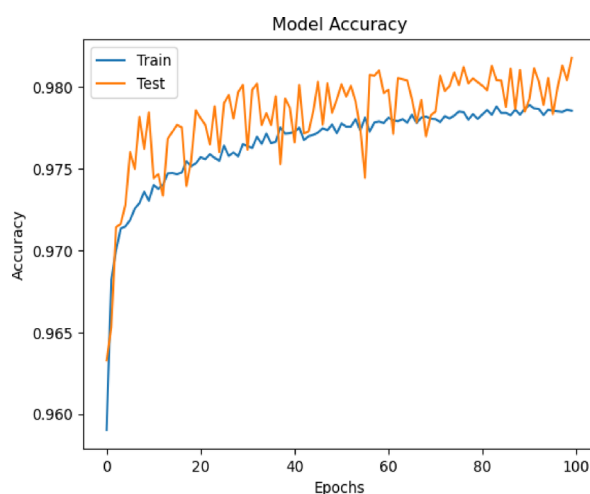


Fig. 5. Learning curve accuracy.

A total of 56,203 test data points were chosen to evaluate the sequential deep learning model. The model correctly predicts 35,666 values for Class 1 and 15,255 values for Class 0. For the understanding of the class distinction, ROC curves of the binary classification model were plotted in Fig. 8. With a variety of threshold values obtained from the probable outcomes of sequential deep learning models, a ROC curve shows the true class rate against the false class rate. The true class rate indicates the likelihood of correctly classifying attacks in contrast to the false class rate, which indicates the possibility of false alarms, which occur when an attack is incorrectly classified.

The confusion matrix shows in Fig. 7 a high number of correct predictions (15255 for class 0 and 35666 for class 1), with relatively fewer misclassifications (1466 false positives and 216 false negatives). Figure 7 shows the model's performance at identifying normal versus malicious traffic through the ROC curve. A high Area under the Curve value in ROC indicates how well the model detects intrusions. This performance achieves an effective relationship between false negatives (missed attacks) and false positives (flagging legitimate traffic). Manufacturers can incorporate this model into their real-time network monitoring programs that operate in critical industries, including banking and healthcare, to minimize financial and privacy exposure from security breaches. The reliable performance of the ROC curve generates confidence in the operational stability of the system.

The ROC curve shows the performance of a binary classification model for both classes, with the true class rate plotted against the false class rate. Both Class 0 and Class 1 curves have an area under the curve (AUC) of 0.99, indicating excellent model performance with high true positive rates and low false positive rates.

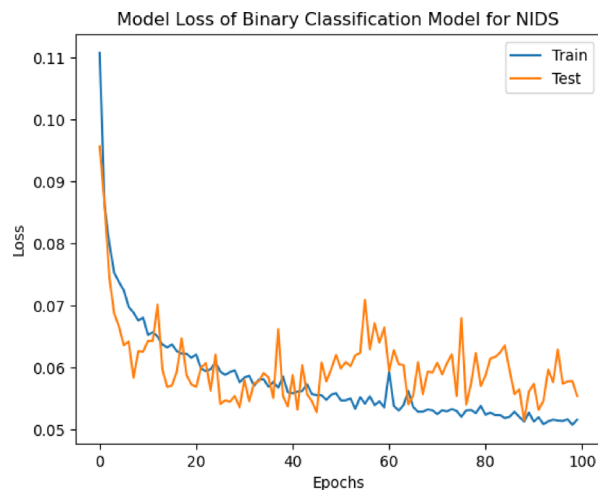


Fig. 6. Model loss.

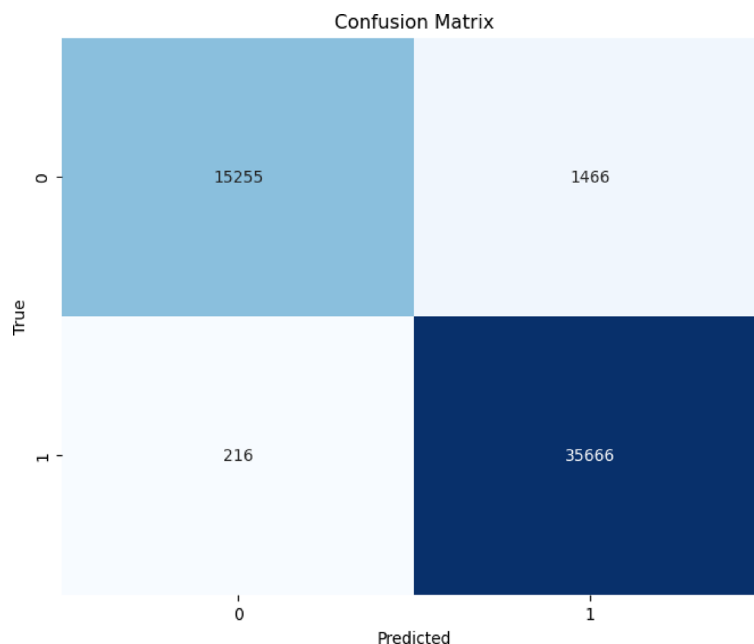


Fig. 7. Confusion matrix for binary classification model.

Comparison with benchmarks

The binary classification models are compared with the benchmark studies^{6,7,37,38,44,45}. Table 7 compares the results of the current study with benchmark cases based on the accuracy score for each model case. The comparison shows that the DNN approach based on the extra tree classifier showed high accuracy in contrast to the other deep learning models.

The comparison of the different models with their accuracy is presented in Fig. 8.

The results of the proposed model, according to Table 7, suggest that the model is the most effective method of intrusion detection. Compared to previous studies (Table 7), the proposed model performed well with the highest accuracy of 97.93% using the DNN model with eight features selected by the Extra Tree Classifier. Moreover, the summarized studies in Table 7 used the sigmoid-based activation function Neural Network Model as compared to the current study, which used the ReLU-based activation function for the DNN model. Hence, the ReLU-based activation function outperforms the other-based activation functions based on the accuracy values.

Discussion

The proposed Sequential Deep Neural Network (DNN) system proved its worth by identifying network intrusions with 97.93% accuracy using a binary classification framework. The model results proved robust and

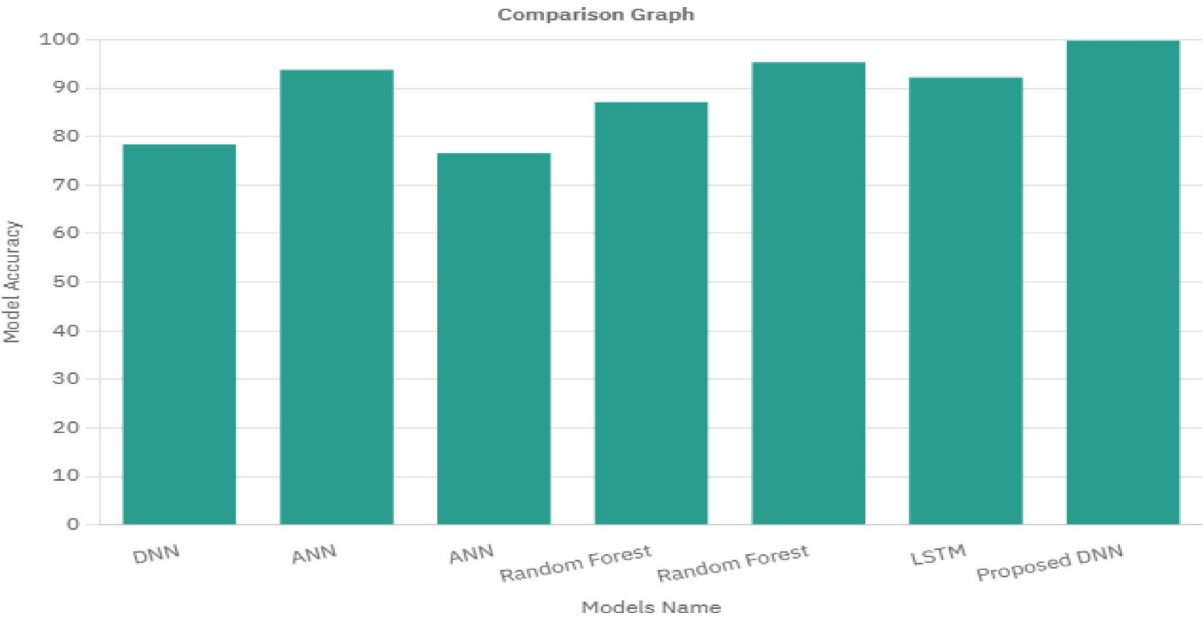


Fig. 8. ROC curve for binary classification model.

References	Model & features	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
44	DNN (43)	78.4	94	73	82
37	ANN (19)	93.75	78.56	92	87.42
45	ANN (43)	79.96	89	65	76
38	Random forest(43)	75.38	83.30	75.38	77.51
6	Random forest (43)	86.	89.5	86	86
7	LSTM(43)	92.2	96	80	87
46	ANN(43)	94	N/A	N/A	N/A
31	Optimized CNN-LSTM (42)	78.47	78	86	82
30	CNN-LSTM (43)	87	85	88	86
Proposed method (sequential deep neural network)		DNN (8)	97.93	97	97

Table 7. Comparison of the proposed model with similar studies.

reliable based on precision, recall, and F1-score performance measurement. The approach used in this study shows better performance than Random Forest, LSTM, and ANNs since it uses L2 regularization, dropout, and ReLU activation to solve overfitting and vanishing gradient problems. The model's performance improved when the Extra Tree Classifier was used to find eight optimal attributes that reduced processing time without dropping accuracy.

The study managed missing data and scaling issues successfully by normalizing essential pre-processing actions. Testing results from confusion matrices and ROC curves reveal the model's reliability through its high AUC, which demonstrates attack versus normal traffic discrimination. The suggested study showed that the model kept performing better in each training iteration while maintaining stable accuracy and loss trends throughout epochs.

The proposed model shows superior performance through a comparative evaluation against benchmark studies. The existing approach solves the previous limitations of unsuitable class imbalance methods, specialized dataset dependency, and inadequate feature selection criteria. Application of the latest UNSW-NB15 dataset helped make the model more relevant because this dataset provides more accurate representations of actual network traffic behavior than KDD-99.

Conclusion

DNN-based models were used in this study to identify network intrusions. The effectiveness of the suggested models was assessed and compared using the UNSW-NB15 dataset. To determine the most important attributes, the suggested models use a variety of pre-processing methods, such as feature selection and data standardization. The results showed that the accuracy of the model was improved by the Extra Tree Classifier's choice of feature selection technique. According to test results, the classification models' performance on the UNSW-NB15

dataset was satisfactory in terms of F1-score metrics, recall, accuracy, and precision. The model outperformed the results of other studies with a binary classification accuracy of 97.93%. The accuracy of the results was demonstrated by the ROC curves and confusion matrices for every class.

Future work

The proposed system provided outstanding results, but additional improvements require attention to reach maximum potential. The model needs expansion to support multiclass classification since it would allow differentiating among network attacks to deliver improved detection capabilities. A comparison of model performance optimization should be evaluated through the application of Recursive Feature Elimination (RFE) and Principal Component Analysis (PCA) as advanced feature selection techniques. The investigation of ensemble learning approaches such as bagging or boosting seeks both accuracy enhancement and decreased false positives during classification tasks. The implementation of transfer learning with pre-trained deep learning models applied to network intrusion detection domains offers both time savings during training and better performance results. Real-time deployment of the system is essential for evaluating its performance rather than artificial scenarios because it allows testing under natural network modifications and altered traffic patterns. The model would work better if we tested different Swish and GELU activation functions alongside AdamW optimization algorithm options. Improved management of these factors will help our system work better at stopping network intrusions. Moreover, the model explainability strategies will be integrated to improve prediction interpretability that allows cybersecurity experts to comprehend the prediction and develop faith in system decisions. Adversarial robustness should also be taken into consideration in the future work to fortify the model against advanced evasion attacks. Lastly, the system could be made more scalable and applicable in the real-world situation by applying federated learning methods to guarantee the privacy and security of data and allowing collaborative education within the decentralized network scenario.

Data availability

The data used to support the findings of this study are available from the corresponding authors upon request.

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Muhammd Amir Khan and Muhammad Farhan perform the Original Writing Part, Software, and Methodology; Hafiz Waheed-ud-din and Tehseen Mazhar perform Rewriting, investigation, design Methodology, and Conceptualization; Saadat Ullah, Umar Farooq Khattak and, Ines Hilali Jaghdam perform the related work part and manage results and discussions; Tehseen Mazhar Muhammd Amir Khan and Muhammad Farhan perform related work part and manage results and discussion; Saadat Ullah, Tehseen Mazhar and Muhammad Amir Khan perform Rewriting, design Methodology, and Visualization; Ines Hilali Jaghdam and Umar Farooq Khattak performs Rewriting, design Methodology, and Visualization.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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